

GUELPH TURFGRASS INSTITUTE, ON, Canada

WMO: 718330

Lat: 43.5472N

Lon: 80.2151W

Elev: 1066

StdP: 14.14

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-6.7	-1.4	-12.6	2.9	-5.3	-7.5	3.8	-0.7	21.1	21.1	19.2	19.7	4.5	70	0.544

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.0	85.5	70.7	82.7	69.4	80.1	68.0	73.6	81.5	71.9	79.3	70.2	77.0	8.3	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
71.1	119.5	77.6	69.5	112.8	75.8	67.8	106.3	74.5	37.9	81.4	36.4	79.6	34.9	77.1	85.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
18.9	16.7	14.9	-14.2	89.3	7.2	3.2	-19.3	91.7	-23.5	93.5	-27.5	95.4	-32.7	97.7		
			-14.0	77.1	7.6	2.9	-19.5	79.2	-23.9	80.8	-28.1	82.5	-33.6	84.5		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	44.1	18.9	20.3	29.2	41.2	53.1	63.1	66.8	65.4	58.7	47.3	36.4	26.7	(6)
(7)		DBStd	18.94	11.58	10.78	11.05	8.55	8.14	6.80	5.65	5.35	7.29	7.97	8.14	9.38	(7)
(8)		HDD50	4090	965	831	653	287	58	1	0	0	10	151	412	722	(8)
(9)		HDD65	7917	1430	1251	1111	714	379	115	46	62	216	549	858	1186	(9)
(10)		CDD50	1919	0	0	7	24	155	394	520	477	270	68	4	0	(10)
(11)		CDD65	271	0	0	0	0	11	58	101	74	26	1	0	0	(11)
(12)		CDH74	2706	0	0	1	8	150	619	970	663	282	13	0	0	(12)
(13)		CDH80	586	0	0	0	0	20	157	226	124	58	1	0	0	(13)
(14)	Wind	WSAvg	6.8	8.1	8.0	8.0	8.2	6.9	5.6	5.2	4.8	5.0	6.3	7.2	7.8	(14)
(15)	Precipitation	PrecAvg	35.6	2.9	2.4	2.4	3.0	3.3	3.3	3.4	3.0	3.1	3.1	3.1	2.7	(15)
(16)		PrecMax	47.0	4.8	4.7	5.5	5.0	6.3	7.7	6.1	6.3	5.6	5.5	6.1	5.0	(16)
(17)		PrecMin	26.4	1.7	1.1	0.8	1.2	1.3	0.9	1.1	0.9	1.0	0.8	1.2	1.3	(17)
(18)		PrecStd	5.1	0.9	0.9	1.1	1.2	1.1	1.6	1.4	1.3	1.2	1.3	1.2	1.0	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	50.7	51.2	69.1	74.5	82.9	87.9	88.7	88.1	86.2	76.0	62.1	54.2	(19)
(20)		MCWB	49.3	47.3	60.2	60.4	66.2	71.7	73.3	71.2	70.9	63.5	54.9	51.8	(20)	
(21)		2%	DB	43.6	43.7	58.1	67.9	78.4	84.5	85.4	83.2	80.8	70.3	57.1	47.7	(21)
(22)		MCWB	41.5	39.9	50.3	56.6	65.1	69.5	71.0	69.7	68.7	62.0	52.8	45.6	(22)	
(23)		5%	DB	38.2	38.2	51.1	62.5	74.0	81.1	82.4	80.3	77.0	65.7	53.9	42.9	(23)
(24)		MCWB	36.6	35.2	46.1	52.6	62.0	68.5	69.5	68.4	66.2	59.0	50.0	41.0	(24)	
(25)		10%	DB	34.0	34.2	44.9	57.1	69.3	77.2	79.5	77.7	72.9	61.5	49.7	38.6	(25)
(26)		MCWB	32.7	32.1	40.3	48.5	58.8	66.2	67.8	67.1	64.5	56.5	46.4	36.9	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	49.5	47.4	60.5	63.8	69.9	74.6	76.7	75.0	73.8	68.4	57.5	51.8	(27)
(28)		MDCB	50.6	50.3	69.1	71.4	79.2	84.2	85.0	82.4	81.4	72.3	60.0	53.1	(28)	
(29)		2%	WB	42.1	40.4	52.0	58.9	67.2	72.1	73.9	72.7	71.0	63.2	53.8	46.1	(29)
(30)		MDCB	43.0	43.3	56.7	65.7	75.5	81.3	81.3	79.9	77.9	67.8	56.1	47.5	(30)	
(31)		5%	WB	36.6	35.1	46.5	54.6	64.3	70.0	72.0	70.9	68.4	60.0	50.0	41.1	(31)
(32)		MDCB	38.2	37.5	51.8	60.7	71.6	77.9	79.0	77.3	74.0	64.6	53.1	42.7	(32)	
(33)		10%	WB	32.7	32.4	40.7	49.2	60.8	68.0	70.0	69.1	65.8	57.0	46.4	37.0	(33)
(34)		MDCB	33.9	34.1	44.6	55.5	67.0	74.8	76.4	75.1	71.1	60.8	50.0	38.5	(34)	
(35)	Mean Daily Temperature Range	MDBR	14.0	15.4	16.7	19.3	21.1	20.7	21.0	21.0	21.4	18.1	14.1	11.7	(35)	
(36)		5% DB	MCDBR	16.5	18.6	25.3	29.7	27.3	25.3	23.9	24.6	26.6	24.8	20.4	16.4	(36)
(37)		MCWBR	15.7	15.9	18.7	19.7	15.6	13.2	12.2	12.9	14.6	16.4	16.1	14.9	(37)	
(38)		5% WB	MCDBR	16.2	17.3	24.3	25.0	24.3	21.6	20.2	19.9	21.4	21.6	18.4	16.0	(38)
(39)		MCWBR	15.9	15.5	19.2	19.0	15.6	12.8	11.9	12.1	13.1	15.7	15.7	15.4	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.286	0.329	0.325	0.363	0.387	0.419	0.410	0.399	0.368	0.350	0.306	0.286	(40)	
(41)		taud	2.371	2.198	2.394	2.334	2.312	2.239	2.281	2.301	2.382	2.418	2.503	2.432	(41)	
(42)		Ebn at Noon	268	270	290	286	281	271	272	271	272	262	261	258	(42)	
(43)		Edh at Noon	25	35	33	38	40	43	41	39	33	28	22	21	(43)	
(44)	All-Sky Solar Radiation	RadAvg	444	705	1108	1408	1704	1875	1869	1633	1309	784	485	331	(44)	
(45)		RadStd	54	82	113	158	169	128	128	83	125	74	65	42	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (51 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page